

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

Paper 2

0862/02

April 2024

1 hour

You must answer on the question paper.

You will need: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 It will take 7 workers 6 days to pick some mangoes.

Calculate how many workers are needed to pick these mangoes in 3 days.

..... [1]

- 2 Calculate.

$$\sqrt{54} \div \sqrt{6}$$

..... [1]

- 3 The mass of a baby is 4 kg.

Each month the mass of the baby increases by 15% of its mass from the previous month.

Find the mass of the baby after 2 months.

..... kg [2]

- 4 In a game Oliver can either lose or draw or win.

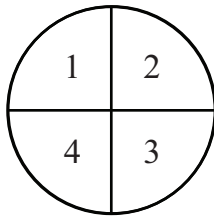
The probability Oliver loses the game is 50%.

The probability Oliver draws the game is 20%.

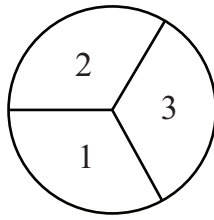
Work out the probability Oliver wins the game.

..... % [1]

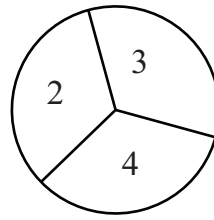
- 5 Anastasia has four fair spinners, A, B, C and D.



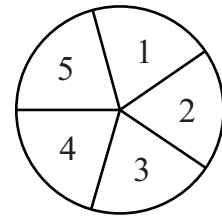
A



B



C



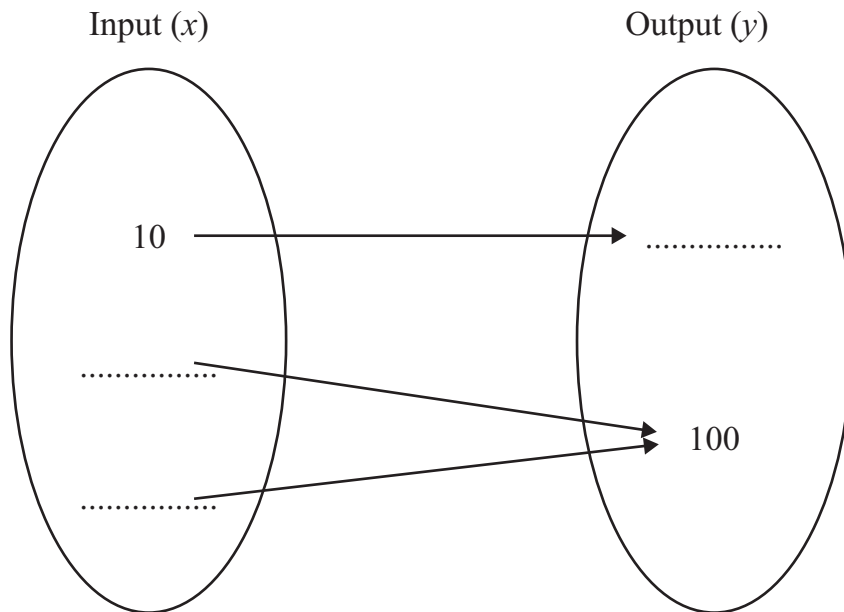
D

She spins one of the spinners 1200 times and gets an even number 486 times.

Write down the letter of the spinner she is most likely to have used.

..... [1]

- 6 Here is a mapping diagram for the function $y = 4x^2$



Complete the mapping diagram with three **different** values.

[3]

7 Here is a calculation.

$$1.5 \times 1.5 \times 28 = \frac{a}{2} \times \frac{a}{2} \times 28 = a \times a \times b$$

Find the value of a and the value of b .

$a =$

$b =$

[2]

8 Carlos sells previously owned clothes.
He will ask his customers one of these questions, A or B.

A On a scale of 1 to 10, what number would you choose to represent the condition of the clothes?

1	2	3	4	5	6	7	8	9	10
Very poor								Excellent	

B Which word(s) would you choose to represent the condition of the clothes?

Very poor	Average	Excellent
-----------	---------	-----------

Carlos wants to work out a **mean** value to represent the condition of the clothes.

Tick (✓) to show which question Carlos should ask his customers and the reason why.

Question A because it asks for quantitative data

☐

Question A because it asks for qualitative data

☐

Question B because it asks for quantitative data

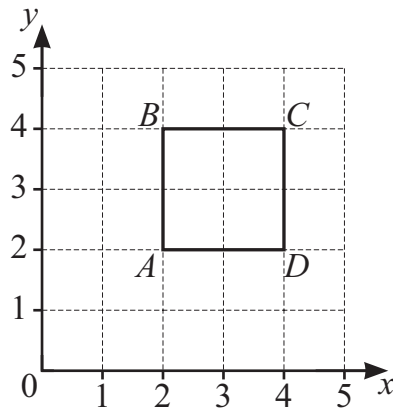
☐

Question B because it asks for qualitative data

☐

[1]

- 9 Yuri enlarges square $ABCD$ by a scale factor of 2



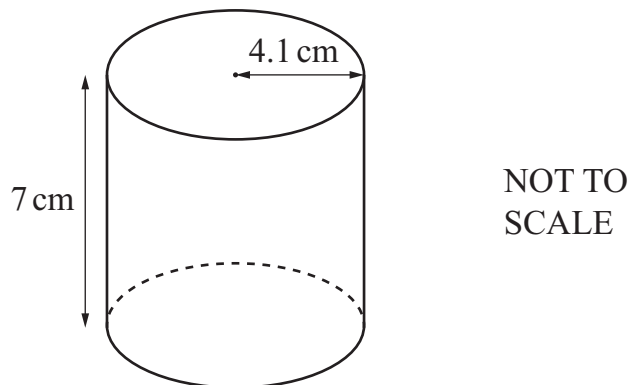
Point A does **not** move when the square is enlarged.

Draw a ring around the coordinates of the centre of enlargement.

(0, 0) (2, 2) (2, 4) (4, 2) (4, 4)

[1]

- 10 The diagram shows a solid metal cylinder with a radius of 4.1 cm and a height of 7 cm.

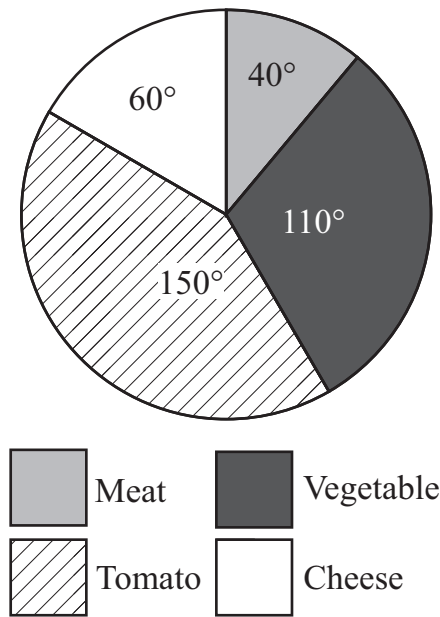


The cylinder is melted and the metal is made into cubes.
The side length of each cube is 2 cm.

Calculate the number of **whole** cubes that are made.

..... [3]

- 11 The pie chart shows information about the different types of pizzas sold in a restaurant.



A total of 324 pizzas are sold.

Mia says, '20 more cheese pizzas are sold than meat pizzas.'

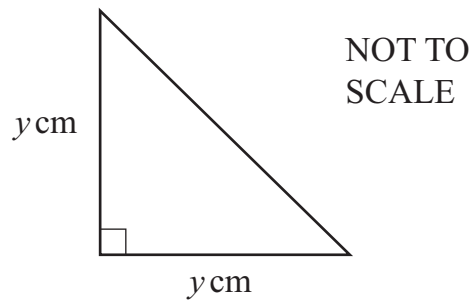
Tick (✓) to show if Mia is correct or not correct.

You must show your working.

Correct ☐ Not correct ☐

[2]

- 12 The diagram shows a right-angled triangle.



Find an expression, in terms of y , for the area of the triangle.

..... cm^2 [1]

- 13 A polygon has 7 sides.

The sizes of the 6 largest interior angles of the polygon add up to 855° .

Calculate the size of the smallest interior angle.

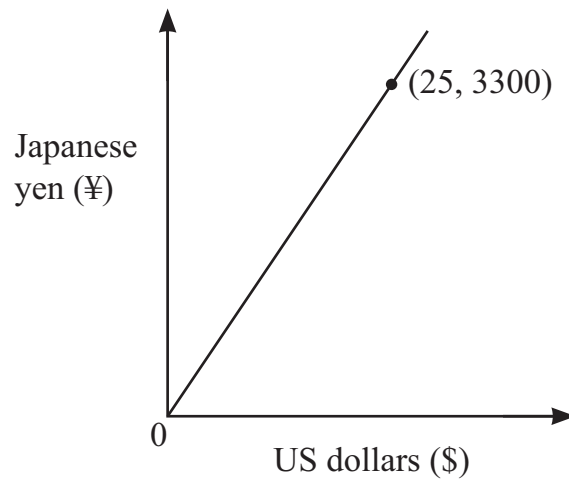
..... $^\circ$ [2]

- 14 Solve.

$$\frac{56}{y+1} = 8$$

$y =$ [2]

- 15 The graph shows the exchange rate between US dollars (\$) and Japanese yen (¥).



Angelique changes \$40 into Japanese yen.

Calculate how many Japanese yen Angelique receives.

¥ [2]

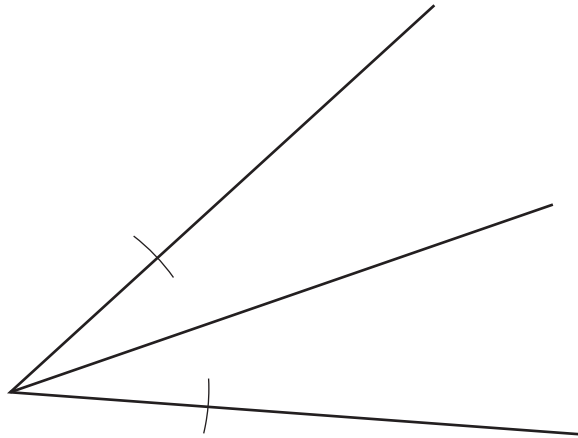
- 16 Triangle ABC is translated by the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ to make triangle DEF .

Then triangle DEF is translated by the vector $\begin{pmatrix} -3 \\ 7 \end{pmatrix}$ to make triangle GHI .

Describe fully the **single** transformation that maps triangle ABC onto triangle GHI .

.....
 [2]

- 17 Rajiv bisects an angle.
Here is his construction.



There are arcs missing from his construction.

Construct the missing arcs accurately on the diagram.

[1]

- 18 The ratio of the sizes of the angles in a triangle is $5 : 8 : 3$

Tick (✓) to show if the triangle is right-angled or not right-angled.
You must show your working.

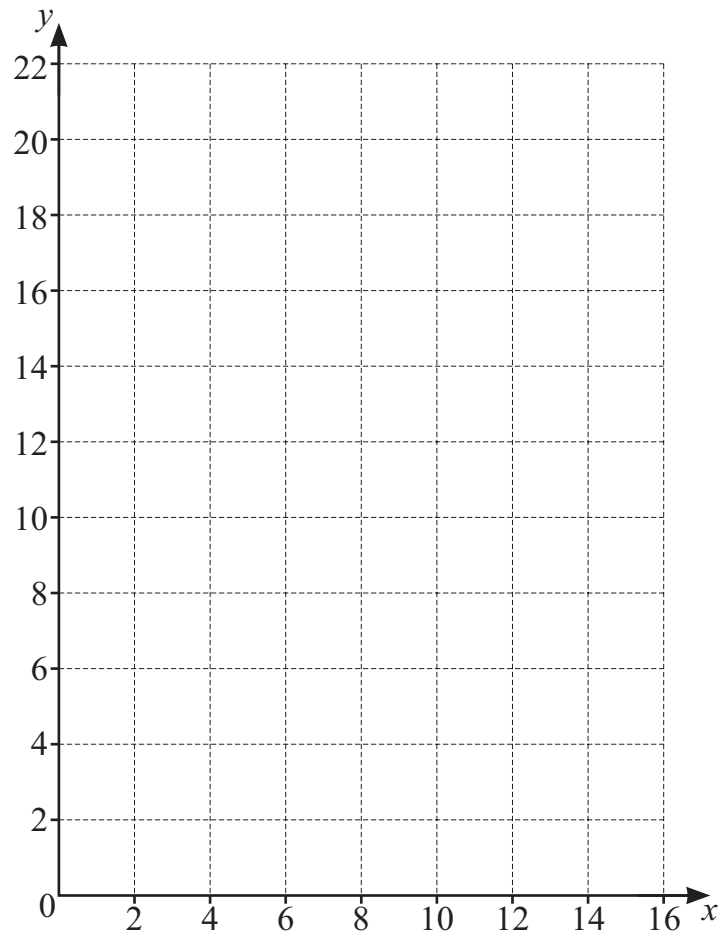
Right-angled ☐ Not right-angled ☐

[2]

19 Complete the table of values for $3y + 5x = 60$

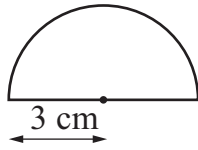
x	0	6	
y			0

On the grid, draw the graph of $3y + 5x = 60$



[3]

- 20 The diagram shows a semicircle with a radius of 3 cm.



NOT TO
SCALE

Tick (✓) to show the area of the semicircle correct to the nearest cm^2 .

14 cm^2 ☐
 19 cm^2 ☐
 28 cm^2 ☐
 57 cm^2 ☐

[1]

- 21 Here is a table of values for points that all lie on the same straight line.

x	5	6	7	11	14
y	27	32	37	57	

Complete the table.

[1]

22 Solve the inequality.

$$2x + 20 \leq 16$$

..... [2]

23 Expand and simplify.

$$x(x + 4) + (x - 3)(x + 5)$$

..... [3]

24 Temperature is measured in $^{\circ}\text{C}$ and in $^{\circ}\text{F}$.

The formula $f = \frac{9c}{5} + 32$ is used to convert c $^{\circ}\text{C}$ to f $^{\circ}\text{F}$.

The approximate formula $f = 2c + 30$ is also used to convert c $^{\circ}\text{C}$ to f $^{\circ}\text{F}$.

Mike says,

‘There is a value of c where these two formulae give an equal value of f .’

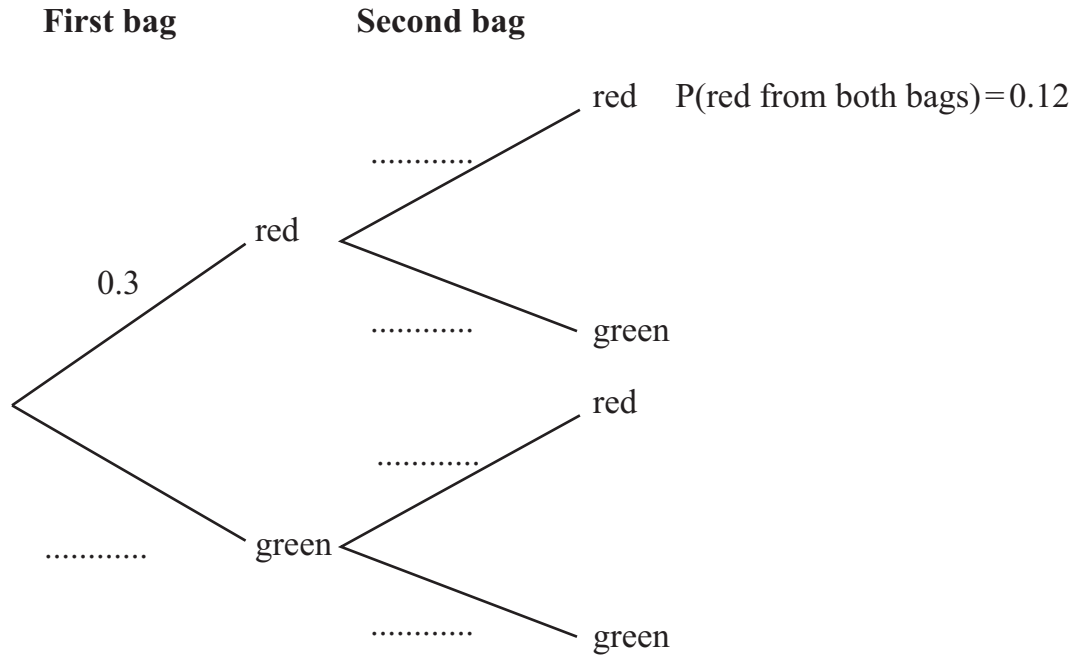
Find the value of c to show that Mike is correct.

$c =$ [3]

- 25** Safia has two bags of sweets.
 Each bag contains red sweets and green sweets only.
 She takes one sweet at random from each bag.

The probability that she takes a red sweet from the first bag is 0.3

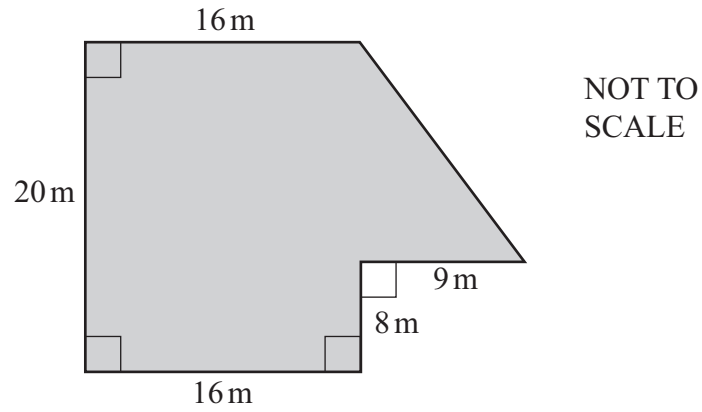
The probability that she takes a red sweet from both bags is 0.12



Complete the **five** missing probabilities on the tree diagram.

[3]

26 The diagram shows a garden in the shape of a hexagon.



Eva builds a fence along all 6 sides of the garden.
The fence costs \$23 per metre.

Calculate the total cost of the fence.

\$ [4]

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